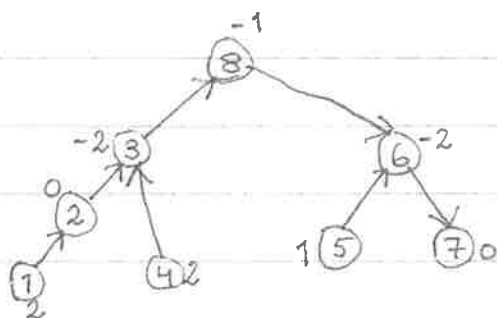


Calculating the tree solution corresponding to a spanning tree

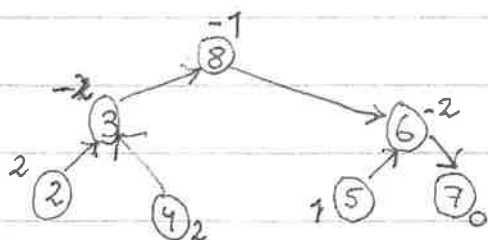
Consider the following spanning tree with directions:



The numbers written next to each node denotes its net supply. Hence $b_1=2, b_2=0, b_3=-2, b_4=2, b_5=1, b_6=-2, b_7=0, b_8=-1$. Observe that $\sum_{i=1}^8 b_i = 0$.

Now consider the leaf ① and the corresponding to ① :

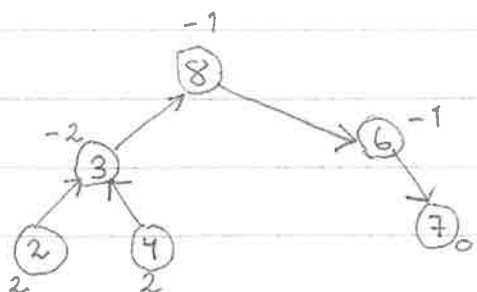
① $\rightarrow$ ② . The supply of ① can only be supplied from ① to ② . Hence we obtain :



$$f_{1,2} = 2$$

Observe that the net supply of ② has been increased by two units since ② receives two units from ①.

We now continue with a leaf of the new graph. We choose ⑤ and the associated arc ⑤ $\rightarrow$ ⑥ . Again the supply of ⑤ can only be sent to ⑥ . Hence we obtain :

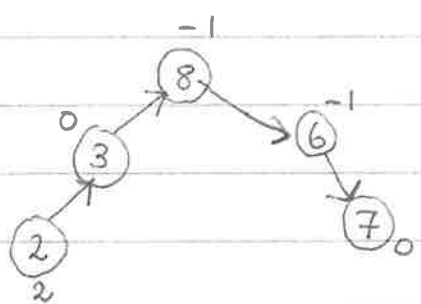


$$f_{1,2} = 2$$

$$f_{5,6} = 1$$

Here the net supply of 6 has changed from (-2) to (-1). This is because 5 has supplied one unit of the demand of 6.

Next considering the leaf 4 and the arc 4 → 3, This gives:

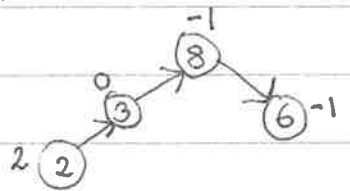


$$f_{1,2} = 2$$

$$f_{5,6} = 1$$

$$f_{4,3} = 2$$

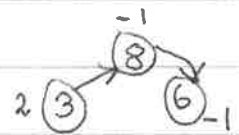
Now consider the leaf 7 and the arc 7 ← 6. We get:



$$f_{1,2} = 2 \quad f_{4,3} = 2$$

$$f_{5,6} = 1 \quad f_{6,7} = 0$$

Next the leaf 2 and the arc 2 → 3 gives:



$$f_{1,2} = 2 \quad f_{4,3} = 2 \quad f_{2,3} = 2$$

$$f_{5,6} = 1 \quad f_{6,7} = 0$$

After the last two steps we obtain:

$$f_{1,2} = 2 \quad f_{4,3} = 2 \quad f_{2,3} = 2 \quad f_{8,6} = 1$$

$$f_{5,6} = 1 \quad f_{6,7} = 0 \quad f_{3,8} = 2$$

which is the basic feasible solution associated with the spanning tree.

A couple of situations did not arise :



Here $\textcircled{a}$ demands 15 units, but there is no supplier. This means -15 must be sent from $\textcircled{a}$ to $\textcircled{b}$, i.e., $f_{a,b} = -15$. Hence this gives a basic infeasible solution.



Here $\textcircled{a}$ supplies 15 units, but place to sent it. This means -15 units must be sent from $\textcircled{b}$ to $\textcircled{a}$, so $f_{b,a} = -15$, and a basic infeasible solution is obtained.